Trust-Aware Adaptive Sampling and Communication for Resource-Constrained IoT Monitoring: A Simulation-Based Evaluation

TAIS Research Project
Author names and affiliations to be supplied before submission

Draft, 19 October 2026

Abstract

Sensor faults can resemble environmental changes, while sparse acquisition can conceal short events. We evaluate a bounded single-node implementation integrating existing sensor-health diagnostics, event-aware schedules, and snapshot emission. This is a simulation-based systems study rather than a claim of a new general adaptive-sampling algorithm. We preserve eight frozen TAIS-related modes and compare six fixed periods, independent delta- and variance-driven samplers, and a clearly labeled Rao-inspired switching control. Parameters and two approximate sampling-budget points are selected on ten development seeds before one held-out evaluation on twenty fresh seeds and all 45 scenarios. Native end-to-end behavior and a common-detector schedule comparison are reported separately. JO-SU-40 achieved 63.00% Recall and 48.90% F1 with 64,488 samples and 655,648 application bytes. Development-selected SU40-budget controls yielded Delta F1 46.08% (sample gap +9.45%); Variance F1 44.02% (sample gap -1.70%); Rao-inspired F1 45.74% (sample gap +1.58%). Individual paired intervals and budget qualification are reported. Delta failed the 5% matching tolerance. All communication counts use actual stateless Binary v1 encoding and independent receiver decoding. Against GateOnly, SU40 used +27.82% samples and produced +7 additional operational fault-induced false reports. The evidence is limited to one synthetic generator, ideal local delivery, and software counters; neither natural-fault robustness nor measured energy savings is established.

1 Introduction

Resource-constrained agricultural monitoring must allocate acquisition and communication effort while retaining useful environmental changes. A sensor may report an out-of-range value, become stuck, drift, or disappear. Responding to every apparent change can waste resources and produce environmental false reports; suppressing questionable readings can discard real events. Sampling and emission introduce distinct losses: an unobserved event cannot be recovered from a correct transmitter, and an observed state transition may remain absent from receiver history if it is not emitted.

Adaptive sampling, fault-aware event detection, predictive transmission, and their coordination have substantial precedents [1–5]. We therefore ask a narrower empirical question: in a reproducible two-channel simulator, what detection, acquisition, communication, and information-age tradeoffs result from integrating existing health diagnostics with causal sampling and explicit emission rules? Are apparent gains retained against independent schedules at nearby acquisition budgets?

The contribution is threefold. First, we expose an executable integration with full business fields, missing-value semantics, and independently decoded receiver edges. Second, we compare frozen internal modes with independent fixed, delta, variance, and literature-inspired schedules under both native and shared-detector views, retaining adverse operating points. Third, we provide seed-level paired uncertainty, budget qualification, counterfactual fault attribution, and an explicit account of observability and modeling limits. These are systems-evaluation contributions. We do not claim first invention of trust-aware sensing, joint sampling and communication, health diagnostics, deadband transmission, or binary compression.

2 Related Work

Adaptive and event-driven acquisition. Rao et al. combine compressed sensing with prediction-based acquisition and adapt switching around critical regions [1]. Masoum et al. exploit correlation and predictions in decentralized quality-aware sampling [2]. DSRA changes acquisition according to signal changes within an adaptive software architecture [6]. These establish both adaptation and architectural integration as prior work. Our Delta and Variance controls are simple independently implemented engineering samplers; they are not reproductions of those papers.

Faults and environmental events. Sharma et al. compare fault-detection families on real deployment datasets [7]; this evidential scope is stronger than injected-fault simulation. Yim and Choi use neighboring readings and confidence to adapt fault-tolerant event decisions [3]. CLARA uses cluster correlation for sampling and reconstructs missing readings at the sink [8]. TAIS has no spatial redundancy or reconstruction of erased environmental truth. Its SensorTrust C11 health rules are an unchanged separately pinned dependency.

Communication and coordinated control. Dias et al. analyze dual prediction and aggregation [4]. Arnaiz et al. coordinate dynamic sampling and dual prediction under energy constraints [5]. Thus coordinated acquisition and communication, multi-mode tradeoffs, and resource objectives are not new concepts here. TAIS emits complete last-sent-deadband snapshots with heartbeat and mandatory state edges. It has no synchronized predictor, network-level optimization, or demonstrated hard energy guarantee. The repository's related-work matrix distinguishes inspected abstracts from detailed method inspection; no cross-paper performance numbers are treated as directly comparable benchmark results.

3 System Architecture

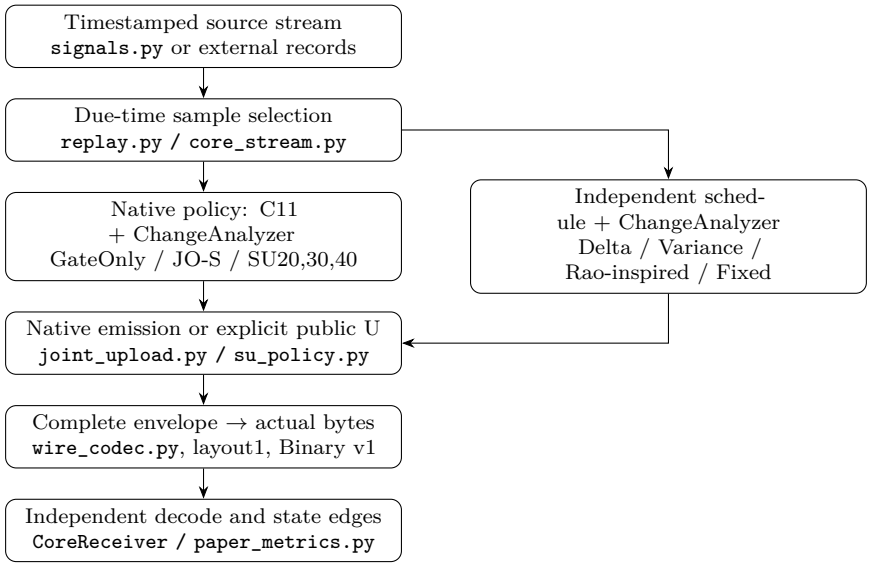


Figure 1: Implemented data path. The common-detector view reuses actual native sample times and replaces detection and emission downstream; it never feeds its outputs back into the schedule. Optional C11 logging on independent methods does not gate their observations.

SensorTrust C11 supplies per-channel state, score, and flags. These are heuristic diagnostics, not calibrated fault probabilities. AdaptiveSense provides the original change analyzer and scheduler. GateOnly uses causal health gating and recovery reseeding without the later confirmation mode. JO-S adds bounded rechecks. JO-SU-20/30/40 retain that sampling design with different idle caps and attach U. U compares current sampled values with the last sent snapshot, emits on event or health state/flag edges and validity changes, applies a two-noise-floor deadband, and has a 120-second heartbeat evaluated at sample times. Health score is retained in every emitted snapshot even when score-only changes do not trigger emission.

Binary v1 preserves timestamps, two values including null, two event booleans, and all available health state/score/flags. We measure encoded byte length, excluding hexadecimal logging representation. Every frame is stateless and independently decodable; CRC checks accidental corruption and provides no authentication. Layout provisioning is shared configuration. There is no ACK, retransmission, packet fragmentation, or deployed LoRaWAN/MQTT stack. The evaluated link delivers every emitted complete frame locally. The receiver only sees decoded frames and detects event starts from received false-to-true transitions. Local cleared states are not implicitly transmitted.

The historical H layer added trend-based health suspicion but lost true events and increased other false reports. It is excluded from the present frozen SU modes; its negative findings remain in the historical evidence rather than being presented as a new successful contribution. GateOnly remains the historical default; this paper does not retroactively reverse prior NO-GO decisions.

4 Methodology

4.1 Synthetic data and split

Each run contains 1,800 one-second source rows with temperature and humidity, a smooth low-amplitude background, independent Gaussian observation noise, and trapezoidal environmental interventions. The generator separates latent environmental values, noisy pre-fault observations, and faulted observations. Labels use physical intervention definitions rather than a method's detection score. Three clean controls and seven fault classes (range, spike, stuck, drift, missing, invalid, stable offset), each crossed with no event, same-channel event, or cross-channel event and short/long profiles, yield 45 scenarios. The actual labeled event window can differ from the injected trapezoid support. Faults affect temperature; cross-channel events affect humidity. This asymmetry is a limitation, not universal multivariate fault coverage.

We audited accessible local structured artifacts, Git-ref histories, and archive seed identities before allocating development seeds 600–609 and final seeds 610–629. Seeds 0–599 were reserved. Fresh seeds change noise and timing jitter within the same generator; they are not external data generalization. All methods share the identical observed trace for each seed/scenario, verified by fingerprints. No external greenhouse record is included in the effectiveness experiment; unresolved source-clock semantics from prior work are not repaired by interpolation.

4.2 Independent schedules and explicit shared components

Fixed5/10/20/30/40/60 reuse the frozen replay implementation, sample the first available source row at or after due time, and upload every sample. On the dense integer grid this gives strict periodic times. They read only due-time values and have no native health diagnostic. The common event analyzer sees only actual sampled points.

For Delta, with channel noise scale σ_c, consecutive finite observations define

$$D_t = \max_c \left\{ \frac{|x_{c,t} - x_{c,t^-}|}{\sigma_c},\ \frac{20|x_{c,t} - x_{c,t^-}|}{\sigma_c(t - t^-)} \right\}. \tag{1}$$

Crossing a development-selected threshold requests the minimum interval. Variance uses elapsed-time EWMA statistics, with $\alpha = 1 - \exp[-(t - t^-)/60]$, residual $d = x_t - m_{t^-}$, $m_t = m_{t^-} + \alpha d$, and $v_t = (1 - \alpha)(v_{t^-} + \alpha d^2)$. It triggers on $\max_c \sqrt{v_{c,t}}/\sigma_c$. Both restore the interval by doubling after 60 seconds of sampled stability, up to the cap. They use constant bounded state; invalidity transitions trigger a dense period without classifying a physical fault.

Rao-inspired uses at most four sampled timestamp/value pairs per channel for least-squares trend fitting and predicts entry into a prior critical band. This is inspired by Rao's line fitting and boundary-time reasoning (their Eqs. 11–13), but is not their full algorithm. Bands are centered on public configured priors (24°C,45%RH), with width set by selected noise-floor multiples. A sampled excursion or a predicted crossing within one cap requests dense observations; an inner 75% band and stable dwell govern return. There is no CS reconstruction, previous-day Student-t tolerance, dual-end prediction, model training/synchronization, or prediction-based substitution. The original paper's performance and guarantees cannot be transferred to this adaptation.

All three native independent samplers use the same sampled-only ChangeAnalyzer for event decisions and upload every sample. Explicit "+C11" versions add unchanged diagnostics without affecting schedules/detection. Three cap 40 "+C11+U" versions add the same TAIS emission component as a diagnostic

ablation and are not completely independent original systems. The *common* view applies identical per-channel ChangeAnalyzer, metadata-only C11, and every-sample emission to every native schedule, including TAIS. It isolates the consequences of sample times, not a complete independent-system comparison. Native TAIS sampling decisions still originate from its native internal health and event states.

4.3 Development selection and budget matching

At cap 40, thresholds 2/4/8 for Delta and Variance and 4/8/12 for Rao-inspired are ranked by development micro F1, then Recall, fewer samples, and smaller threshold. Selected thresholds are 2, 2, and 4, respectively. Native minimum 5 s and caps 20/40/60 remain in the final tradeoff curves. All fixed periods remain.

An initial cap-only development search failed to match either target; those results are retained. A disclosed development-only refinement searches independent minimum intervals 10/15/20/25/30/35s and caps 20–60 s in 5 s steps, requiring minimum $\leq$cap, at the already selected quality thresholds. Among points within 5% of the GateOnly or SU40 development sample total, the rule selects highest F1, then closest budget, larger cap, and smaller minimum; if none qualify, it retains the closest failure. These bounds impose explicit acquisition budgets and can reduce short-event observability. Frozen TAIS points are unchanged. Actual final gaps determine whether a comparison qualifies as approximately matched; final data never adjust parameters. All code/configuration/points were committed before final execution.

4.4 Outcomes and uncertainty

The frozen one-to-one channel-aware event matcher uses a 5 s tolerance. A report may therefore precede a label boundary by up to 5 s; this tolerance does not imply forecasting or future access. We report TP/FP/FN and derived Precision/Recall/F1 separately for sender-on-upload and receiver-edge reports, with local detection counts also retained. Recall is undefined, not zero, in no-event runs. Short/long and same/cross groups remain separate. Conditional event delays must be read together with misses; matched-in-both delay differences are separately available. The operational "fault-induced" count is an unmatched fault-channel report within the fault or 120 s recovery window that lacks a coincident report (within 5 s) in the same-policy clean counterfactual. This is a simulator attribution definition, not proof of physical causal diagnosis.

Costs count actual sampled cycles, uploads, encoded bytes, mean bytes/upload, source-second pooled P95/maximum age, and sample-to-emission delay for exact sampled event/health signatures. Physical-event-to-report delay is distinct. No receiver carries a future value backward in time, and decoded null remains null. Health detection uses raw C11 flags; capability and rate are N/A for methods without health metadata.

We resample whole random seeds in paired fashion (5,000 bootstrap draws, fixed RNG2026101917); scenario rows and source seconds within a seed are not independent replicates. Intervals describe uncertainty within the synthetic family, and many contrasts/endpoints prohibit a broad significance claim from one favorable comparison. All per-run metrics, grouped summaries, candidate results, fingerprints, and frame archives are retained or deterministically reproducible.

5 Experimental Results

The final evaluation contains 73,800 scored method/view runs on 900 seed–scenario streams, with 600 labeled events per operating point. Table 1 separates the costs of native behavior. Compared with AS+C11, SU40 changed Recall from 50.17% to 63.00%, F1 from 38.32% to 48.90%, samples by -4.07%, and bytes by -43.92%. This comparison alone is insufficient: against GateOnly the F1 difference is +8.32 points, 95% CI [+4.01, +12.40], accompanied by +27.82% samples and +7 fault-induced reports. Fixed competitors and all adverse modes remain visible.

5.1 Nearby acquisition budgets

5/6 native independent budget points remained within 5% in the held-out split. No parameters were adjusted. Gate versus Delta-bGate+C11: final sample gap +1.22%, target-minus-independent F1 -2.30 points, 95% CI [-6.24, +1.52]; qualifies as approximately matched. SU40 versus Delta-bS40+C11: final sample gap +9.45%, target-minus-independent F1 +2.82 points, 95% CI [-0.11, +5.43]; does not qualify as matched.

Table 1: Representative native modes on the final split. P/R/F1 are sender percentages; FI is the operational fault-induced false-report count. "AS" lacks health information. Every other listed mode carries full C11 metadata. Full native operating points appear in the appendix. Var and Rao abbreviate Variance and Rao-inspired; S denotes JO-S.

Method	P	R	F1	Samples	Bytes	FI
AS	31.00	50.17	38.32	67,222	969,950	447
AS+C11	31.00	50.17	38.32	67,222	1,169,204	447
Gate	34.72	48.83	40.58	50,453	1,113,522	306
Gate+U	34.72	48.83	40.58	50,453	617,988	306
S	43.06	71.83	53.84	100,810	1,241,098	319
SU20	43.06	71.83	53.84	100,810	699,046	319
SU30	39.10	61.00	47.66	76,736	665,780	316
SU40	39.96	63.00	48.90	64,488	655,648	313
Fixed-5+C11	36.57	72.17	48.54	324,000	11,418,840	540
Fixed-10+C11	35.81	69.83	47.34	162,000	5,709,420	551
Fixed-20+C11	25.19	44.50	32.17	81,000	2,854,260	441
Fixed-30+C11	22.12	35.17	27.16	54,000	1,902,540	418
Fixed-40+C11	28.40	42.50	34.05	40,500	1,426,680	385
Fixed-60+C11	31.06	43.17	36.12	27,000	950,820	371
Delta-40+C11	36.94	68.83	48.08	215,098	7,528,786	477
Var-40+C11	35.76	66.33	46.47	119,476	4,154,304	468
Rao-40+C11	35.96	67.00	46.80	106,051	3,674,796	470

Gate versus Var-bGate+C11: final sample gap +4.64%, target-minus-independent F1 +1.10 points, 95% CI [-3.54, +5.76]; qualifies as approximately matched. SU40 versus Var-bS40+C11: final sample gap -1.70%, target-minus-independent F1 +4.88 points, 95% CI [+0.82, +8.84]; qualifies as approximately matched. Gate versus Rao-bGate+C11: final sample gap -0.69%, target-minus-independent F1 -3.23 points, 95% CI [-4.92, -1.56]; qualifies as approximately matched. SU40 versus Rao-bS40+C11: final sample gap +1.58%, target-minus-independent F1 +3.17 points, 95% CI [-0.49, +6.61]; qualifies as approximately matched. The

Table 2: Development-selected independent +C11 budget points. Differences are TAIS target minus independent method in percentage points; intervals are descriptive seed-paired95% F1 intervals. Gap is independent samples relative to target. Bounds are minimum/cap in seconds. Qualification is based on final actual samples.

Target	Independent+C11	Min/cap	Gap	ΔR	ΔF1	F1 CI	Within 5%
Gate	Delta-bGate	15/40	+1.22%	-9.17	-2.30	[-6.24, +1.52]	yes
SU40	Delta-bS40	10/50	+9.45%	-0.67	+2.82	[-0.11, +5.43]	no
Gate	Var-bGate	15/50	+4.64%	-4.50	+1.10	[-3.54, +5.76]	yes
SU40	Var-bS40	10/60	-1.70%	+2.83	+4.88	[+0.82, +8.84]	yes
Gate	Rao-bGate	10/60	-0.69%	-11.33	-3.23	[-4.92, -1.56]	yes
SU40	Rao-bS40	10/35	+1.58%	+0.00	+3.17	[-0.49, +6.61]	yes

complete table includes comparisons that favor independent methods. Higher Recall at higher sample totals is not evidence of equal-budget superiority. The common-detector view is reported in Figure 3 and the appendix; differences from native outcomes show the contribution of gating and state resets as well as scheduling.

5.2 Faults, short events, and receiver state

Table 3 includes every fault class. For drift, GateOnly/SU40 Recall is 73.75%/88.75%, and operational induced counts are 177/182. Stable-offset groups yield Recall 97.50%/92.50%. These failures are not removed from overall totals. Short-event Recall is 25.00% for GateOnly and 54.67% for SU40; long-event rates are separately tabulated. Gate: sender TP/FP/FN 293/551/307, receiver 293/550/307; receiver F1 40.61%; sampled important edges unreported 1, maximum reported edge delay 55s. Gate+U: sender TP/FP/FN 293/551/307, receiver 293/551/307; receiver F1 40.58%; sampled important edges unreported 0, maximum reported edge delay 0s. SU20: sender TP/FP/FN 431/570/169, receiver 431/570/169; receiver F1 53.84%; sampled important edges unreported 0, maximum reported edge delay 0s. SU40: sender TP/FP/FN 378/568/222, receiver 378/568/222; receiver F1 48.90%; sampled important edges unreported 0, maximum reported edge delay 0s. All corresponding local detections, separate event/health edge delays, uploads, mean bytes, receiver age and reconstruction coverage remain in the complete machine-readable summaries.

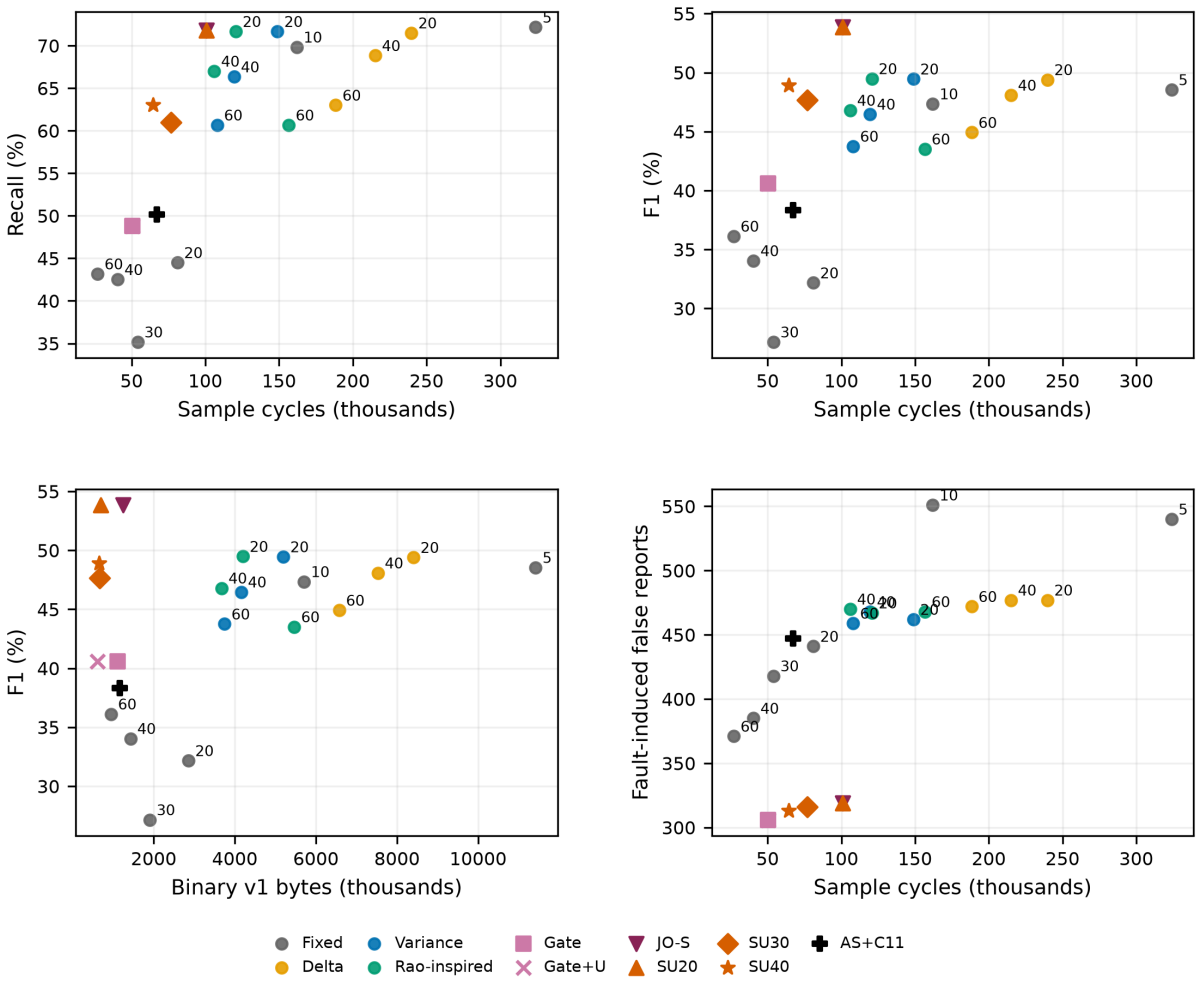


Figure 2: Native operating points with full health metadata. Labels on independent/fixed points denote caps/periods; their native minimum is 5 s. Points are discrete, with no smoothing or interpolated operating points. Budget-bound variants are tabulated separately.

Table 3: All fault classes, retaining adverse groups. R is sender Recall (%); FI is operational fault-induced reports. No-event runs remain in false-report totals; their individual Recall is N/A.

Fault	Gate R	SU40 R	Fixed30 R	Gate FI	SU40 FI	Fixed30 FI
clean	57.50	85.00	50.00	0	0	0
range	27.50	41.25	35.00	0	0	172
spike	56.25	83.75	48.75	5	4	6
stuck	27.50	41.25	25.00	0	0	0
drift	73.75	88.75	50.00	177	182	122
missing	27.50	41.25	25.00	0	0	0
invalid	27.50	41.25	25.00	0	0	0
stable_offset	97.50	92.50	30.00	124	127	118

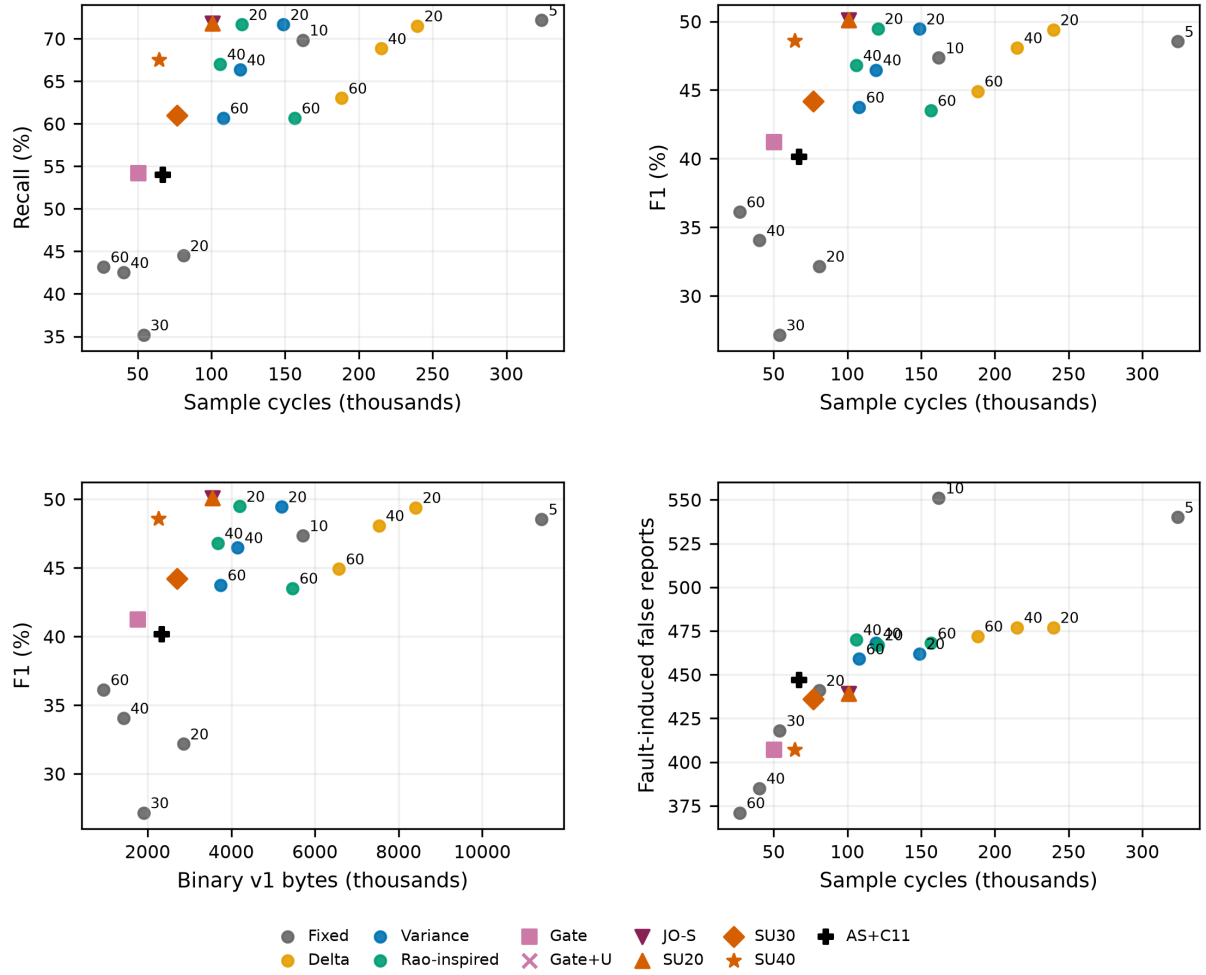


Figure 3: Actual native sample schedules under identical detection, C11 logging and every-sample emission. This is schedule isolation. It is not a claim that external original methods contain TAIS health gates.

Table 4: Selected mode boundaries. Short/long/same/cross are sender Recall percentages. FD is raw C11 any-flag detection rate; event delay is conditional on sender hits. P95/max age pools all source seconds for that method.

Mode	Short	Long	Same	Cross	FD	Delay(s)	Age(s)
Gate	25.00	72.67	35.62	63.93	54.05	28.53	58/79
Gate+U	25.00	72.67	35.62	63.93	54.05	28.53	115/159
SU20	71.33	72.33	48.44	98.57	62.62	15.29	113/134
SU40	54.67	71.33	43.12	85.71	54.88	16.80	113/154
Fixed-20+C11	22.33	66.67	28.75	62.50	53.10	26.48	18/19
Fixed-30+C11	1.33	69.00	22.19	50.00	51.19	42.19	28/29
Delta-bS40+C11	57.00	70.33	40.62	90.00	52.26	17.93	45/49
Var-bS40+C11	47.00	73.33	39.06	84.29	51.55	19.94	55/59
Rao-bS40+C11	56.00	70.00	39.38	90.00	50.00	17.27	32/34

Table 5: Communication and exact sampled-edge delivery. E/H denote event/health. Edge lags concern detected sampled transitions, not physical occurrence. U rows on independent methods explicitly reuse a TAIS component.

Mode	Tx	B/Tx	Age P95/max(s)	Missed	Lag E/H(s)
AS+C11	33,209	35.21	57/79	58	55/55
Gate	31,645	35.19	58/79	1	55/0
Gate+U	17,575	35.16	115/159	0	0/0
S	35,317	35.14	56/59	0	55/0
SU20	19,921	35.09	113/134	0	0/0
SU30	18,952	35.13	114/144	0	0/0
SU40	18,687	35.09	113/154	0	0/0
Delta-40+C11+U	22,220	35.05	112/129	0	0/0
Var-40+C11+U	20,909	35.01	113/149	0	0/0
Rao-40+C11+U	20,908	35.02	113/149	0	0/0

5.3 Conditional resource sensitivity

$$C = N_{\text{sample}}a + N_{\text{upload}}b + B_{\text{v1}} + P. \tag{2}$$

Byte cost is normalized to1; sample-to-byte and fixed-upload-to-byte ratios vary independently over 0.01–10,000. Figure 4 assumes equal processing cost (P cancels), not measured equivalence. The accompanying grid also gives the extra-processing break-even allowance. No device energy coefficients are inferred. These plots illustrate conditional software-cost rankings and cannot support battery-life percentages.

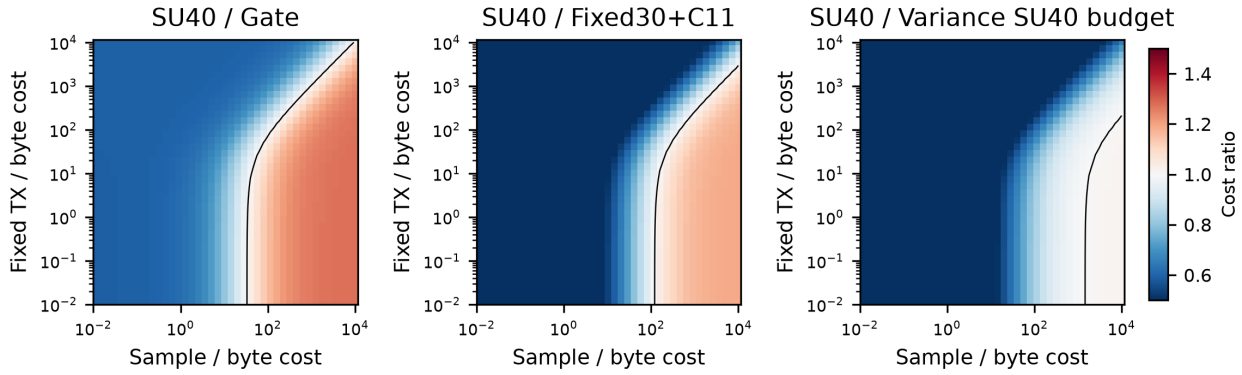


Figure 4: SU40 modeled cost divided by each named comparator. Ratios below1 favor SU40 only under the displayed coefficients and equal-processing assumption. No current, radio airtime or joules were measured.

6 Discussion

Adding U to Delta-40+C11 preserved sender TP/FP and sample counts while changing bytes by -89.66% and pooled age P95 from 34s to 112s; important missed edges were 0. Adding U to Var-40+C11 preserved sender TP/FP and sample counts while changing bytes by -82.38% and pooled age P95 from 37s to 113s; important missed edges were 0. Adding U to Rao-40+C11 preserved sender TP/FP and sample counts while changing bytes by -80.08% and pooled age P95 from 37s to 113s; important missed edges were 0. In the common-detector view SU40 F1 was 48.59%, compared with its native 48.90%. The difference is a detector/gating comparison under the same sample times, not a new schedule. At the lower Gate budget, Rao-inspired F1 was 43.81% versus GateOnly 40.58%; Gate-minus-Rao F1 CI [-4.92, -1.56]. This adverse quality comparison remains despite higher independent-method fault-induced reporting. GateOnly is not universally optimal. Gate+U also restored a received false edge: receiver false reports changed from 550 to 551, although sender reports were unchanged. Complete edge delivery does not necessarily improve receiver F1. More frequent observations can increase short-event coverage and allow a debounced detector to confirm events before they disappear. They also expose more sensor artifacts, and the health detector itself receives a different sequence. Consequently, acquisition counts and fault-induced false reports must

accompany Recall. A sampled raw health flag can be wrong or delayed; it is not evidence of a uniquely identified physical failure.

Emission U changes a different objective: it can preserve sampled event and health edges while reducing ordinary snapshots, thereby increasing normal-data age. Its diagnostic use on independent samplers is important: any similar gain there is evidence for a reusable communication rule, not exclusive novelty of the TAIS schedule. Joint sampling and communication effects must not be inferred by deleting health fields or comparing binary bytes against JSON on a competitor.

The historical H failure warns against interpreting a trend as a physical fault: slow environmental changes and stable offsets can resemble sensor drift, and added suspicion can suppress true events. Historical SU20 prioritizes detection with extra sampling; SU40 is a different resource operating point, not a universal improvement. The present comparison retains both, as well as GateOnly and the simpler GateOnly+U option. The study offers evidence for choosing a mode under explicit objectives, not a theorem that all objectives can be optimized simultaneously.

7 Limitations and Threats to Validity

Observability and identifiability. An event entirely between samples is unavailable to every sampled-only method. A fault that completely overwrites the true same-channel signal removes recoverable information. Without independent references, stable offsets and some drift/environmental-change combinations are ambiguous. More samples cannot reconstruct information that is absent from the sensor output. Greater Recall can coexist with greater fault-induced false reporting.

Synthetic scope and tuning. All main outcomes come from one generator with fixed profile families, two channels, temperature-only fault injection, a narrow event-start distribution and bounded timing jitter. Seeds are paired noise/timing replicates, not independent farms or sensor technologies. Rao-inspired assumes public prior centers that coincide with the configured generator baselines; those priors would require application knowledge in a new environment. The two-stage threshold and interval-bound grid is not an exhaustive search for each independent algorithm. Development threshold and budget searches introduce selection within that family; uncertainty intervals do not account for model misspecification. The extra bound search was disclosed during development, before any held-out test, and does not establish general optimality. No natural-fault labels or agricultural field events were evaluated. Previously obtained greenhouse files are not used to invent a source time axis.

Baseline and implementation scope. Delta and Variance are original lightweight engineering controls, not state-of-the-art literature reproductions. Rao-inspired omits major components of Rao's method. Shared ChangeAnalyzer limits detector diversity, and public-component variants must not be confused with external original methods. A faithful suitable literature reproduction and a second independently specified signal/fault family would materially strengthen external claims. The current systems-study framing remains narrower than an algorithm-innovation claim.

Transport, hardware, and metrics. The main link is ideal local byte delivery. Application bytes exclude radio headers, receive windows, contention, spreading factor effects, loss and retries. LoRa airtime is modulation- and packet-dependent; neither byte ratios nor the additive sensitivity model measure actual wireless energy. There is no MCU timing, RAM characterization, current measurement or battery-lifetime validation. A sample cycle is a software processing opportunity, not measured sensor power gating. Conditional delays exclude misses; fault-induced attribution depends on the selected counterfactual coincidence window. The label matcher and channel asymmetry may favor some event geometries. Multiple contrasts remain descriptive, and no global field effectiveness is established.

8 Conclusion

The held-out results show a conditional operating tradeoff: SU40 F1 was 48.90%, with 64,488 samples and 655,648 bytes, while GateOnly F1 was 40.58%. Of the three SU40-targeted independent controls, 2 qualified within 5% on final samples. Against Variance, the F1 difference was +4.88 points, 95% CI [+0.82, +8.84]. Against Rao-inspired, the F1 difference was +3.17 points, 95% CI [-0.49, +6.61]. Delta had a +9.45% sample

gap and cannot support a matched-budget superiority claim. Full native and common-view comparisons, adverse lower-budget results and interval uncertainty prevent a universal-superiority conclusion. TAIS is best understood here as an auditable prototype and design/tradeoff study. The integration enables explicit sender/receiver analysis and conditional resource comparisons, but known techniques and synthetic evidence do not justify a claim of a universally superior new sensing algorithm. Independent near-budget controls, adverse cases, and information-age costs are necessary parts of the conclusion.

Artifact and attribution statement

Source, exact branch, freeze commit, complete tables, configuration and one-command reproduction are documented at `https://github.com/Sver0411/TAIS/tree/research/tais-paper-evaluation`. AdaptiveSense commit`f7ec30fd` and SensorTrust commit`2ce98dd0` are preserved unmodified. Manuscript tables and numerical paragraphs are generated from final CSV files, with a cell-level audit CSV. Full compressed traces remain in an ignored local archive and can be regenerated; committed manifests enumerate hashes. This draft has not undergone peer review and makes no publication guarantee.

References

[1] Yuan Rao, Gang Zhao, Wen Wang, Jingyao Zhang, Zhaohui Jiang, and Ruchuan Wang. Adaptive data acquisition with energy efficiency and critical-sensing guarantee for wireless sensor networks. *Sensors*, 19(12):2654, 2019. doi: 10.3390/s19122654.

[2] Alireza Masoum, Nirvana Meratnia, and Paul J. M. Havinga. A decentralized quality aware adaptive sampling strategy in wireless sensor networks. In *2012 9th International Conference on Ubiquitous Intelligence and Computing and 9th International Conference on Autonomic and Trusted Computing*, pages 298–305, 2012. doi: 10.1109/UIC-ATC.2012.156.

[3] Sung-Jib Yim and Yoon-Hwa Choi. An adaptive fault-tolerant event detection scheme for wireless sensor networks. *Sensors*, 10(3):2332–2347, 2010. doi: 10.3390/s100302332.

[4] Gabriel Martins Dias, Boris Bellalta, and Simon Oechsner. The impact of dual prediction schemes on the reduction of the number of transmissions in sensor networks. *Computer Communications*, 112:58–72, 2017. doi: 10.1016/j.comcom.2017.08.002.

[5] David Arnaiz, Francesc Moll, Eduard Alarcón, and Xavier Vilajosana. Energy and relevance-aware adaptive monitoring method for wireless sensor nodes with hard energy constraints. *Integration*, 94: 102097, 2024. doi: 10.1016/j.vlsi.2023.102097.

[6] Hatem Algabroun. Dynamic sampling rate algorithm (dsra) implemented in self-adaptive software architecture: a way to reduce the energy consumption of wireless sensors through event-based sampling. *Microsystem Technologies*, 26(4):1067–1074, 2020. doi: 10.1007/s00542-019-04631-9.

[7] Abhishek B. Sharma, Leana Golubchik, and Ramesh Govindan. Sensor faults: Detection methods and prevalence in real-world datasets. *ACM Transactions on Sensor Networks*, 6(3):1–39, 2010. doi: 10.1145/1754414.1754419.

[8] Hassan Harb, Clara Abou Nader, Ali Jaber, Mourad Hakem, Jean-Claude Charr, Chady Abou Jaoude, and Chamseddine Zaki. Clara: A cluster-based node correlation for sampling rate adaptation and fault tolerance in sensor networks. *Internet of Things*, 28:101345, 2024. doi: 10.1016/j.iot.2024.101345.

A Complete native operating points

Common analyzer and per-sample emission apply to independent native schedules. +C11 adds health metadata only; +C11+U adds the public TAIS upload component. Health-free native payloads are functionally unequal. Budget names denote development-selected points, not guaranteed equality. All45 scenarios and all20 final seeds are included in each row.

Method	P	R	F1	RxF1	Samples	Bytes	FI
AS	31.00	50.17	38.32	38.34	67,222	969,950	447
AS+C11	31.00	50.17	38.32	38.34	67,222	1,169,204	447
Delta-bGate	34.02	58.00	42.88	42.88	51,070	1,486,276	441
Delta-bGate+C11	34.02	58.00	42.88	42.88	51,070	1,792,696	441
Delta-bS40	36.11	63.67	46.08	46.08	70,584	2,050,822	467
Delta-bS40+C11	36.11	63.67	46.08	46.08	70,584	2,474,326	467
Delta-20	37.73	71.50	49.40	49.40	239,766	6,970,632	477
Delta-20+C11	37.73	71.50	49.40	49.40	239,766	8,409,228	477
Delta-40	36.94	68.83	48.08	48.08	215,098	6,238,198	477
Delta-40+C11	36.94	68.83	48.08	48.08	215,098	7,528,786	477
Delta-40+C11+U	36.94	68.83	48.08	48.08	215,098	778,846	477
Delta-60	34.90	63.00	44.92	44.92	188,337	5,443,733	472
Delta-60+C11	34.90	63.00	44.92	44.92	188,337	6,573,755	472
Fixed-10	35.81	69.83	47.34	47.34	162,000	4,737,420	551
Fixed-10+C11	35.81	69.83	47.34	47.34	162,000	5,709,420	551
Fixed-20	25.19	44.50	32.17	32.17	81,000	2,368,260	441
Fixed-20+C11	25.19	44.50	32.17	32.17	81,000	2,854,260	441
Fixed-30	22.12	35.17	27.16	27.16	54,000	1,578,540	418
Fixed-30+C11	22.12	35.17	27.16	27.16	54,000	1,902,540	418
Fixed-40	28.40	42.50	34.05	34.05	40,500	1,183,680	385
Fixed-40+C11	28.40	42.50	34.05	34.05	40,500	1,426,680	385
Fixed-5	36.57	72.17	48.54	48.54	324,000	9,474,840	540
Fixed-5+C11	36.57	72.17	48.54	48.54	324,000	11,418,840	540
Fixed-60	31.06	43.17	36.12	36.12	27,000	788,820	371
Fixed-60+C11	31.06	43.17	36.12	36.12	27,000	950,820	371
Gate	34.72	48.83	40.58	40.61	50,453	1,113,522	306
Gate+U	34.72	48.83	40.58	40.58	50,453	617,988	306
S	43.06	71.83	53.84	53.84	100,810	1,241,098	319
SU20	43.06	71.83	53.84	53.84	100,810	699,046	319
SU30	39.10	61.00	47.66	47.66	76,736	665,780	316
SU40	39.96	63.00	48.90	48.90	64,488	655,648	313
Rao-bGate	34.45	60.17	43.81	43.81	50,105	1,432,818	462
Rao-bGate+C11	34.45	60.17	43.81	43.81	50,105	1,733,448	462
Rao-bS40	35.90	63.00	45.74	45.74	65,510	1,889,669	466
Rao-bS40+C11	35.90	63.00	45.74	45.74	65,510	2,282,729	466
Rao-20	37.79	71.67	49.48	49.48	120,666	3,473,344	467
Rao-20+C11	37.79	71.67	49.48	49.48	120,666	4,197,340	467
Rao-40	35.96	67.00	46.80	46.80	106,051	3,038,490	470
Rao-40+C11	35.96	67.00	46.80	46.80	106,051	3,674,796	470
Rao-40+C11+U	35.96	67.00	46.80	46.80	106,051	732,144	470
Rao-60	33.92	60.67	43.51	43.51	156,591	4,527,190	468
Rao-60+C11	33.92	60.67	43.51	43.51	156,591	5,466,736	468
Var-bGate	31.34	53.33	39.48	39.48	52,795	1,528,966	450
Var-bGate+C11	31.34	53.33	39.48	39.48	52,795	1,845,736	450
Var-bS40	34.71	60.17	44.02	44.02	63,391	1,826,182	454
Var-bS40+C11	34.71	60.17	44.02	44.02	63,391	2,206,528	454
Var-20	37.75	71.67	49.45	49.45	148,887	4,305,526	462
Var-20+C11	37.75	71.67	49.45	49.45	148,887	5,198,848	462
Var-40	35.76	66.33	46.47	46.47	119,476	3,437,448	468
Var-40+C11	35.76	66.33	46.47	46.47	119,476	4,154,304	468
Var-40+C11+U	35.76	66.33	46.47	46.47	119,476	731,940	468
Var-60	34.21	60.67	43.75	43.75	108,040	3,099,572	459
Var-60+C11	34.21	60.67	43.75	43.75	108,040	3,747,812	459

B Complete common-detector schedules

Every row uses full C11 metadata and every-sample emission, with no health gating in this view. Different TAIS native schedules can be identical (e.g., with and without U); duplicate outcomes are retained for explicit ablation.

Method	P	R	F1	RxF1	Samples	Bytes	FI
AS	31.95	54.00	40.15	40.15	67,222	2,342,578	447
AS+C11	31.95	54.00	40.15	40.15	67,222	2,342,578	447
Delta-bGate	34.02	58.00	42.88	42.88	51,070	1,792,696	441
Delta-bS40	36.11	63.67	46.08	46.08	70,584	2,474,326	467
Delta-20	37.73	71.50	49.40	49.40	239,766	8,409,228	477
Delta-40	36.94	68.83	48.08	48.08	215,098	7,528,786	477
Delta-60	34.90	63.00	44.92	44.92	188,337	6,573,755	472
Fixed-10	35.81	69.83	47.34	47.34	162,000	5,709,420	551
Fixed-20	25.19	44.50	32.17	32.17	81,000	2,854,260	441
Fixed-30	22.12	35.17	27.16	27.16	54,000	1,902,540	418
Fixed-40	28.40	42.50	34.05	34.05	40,500	1,426,680	385
Fixed-5	36.57	72.17	48.54	48.54	324,000	11,418,840	540
Fixed-60	31.06	43.17	36.12	36.12	27,000	950,820	371
Gate	33.27	54.17	41.22	41.22	50,453	1,771,306	407
Gate+U	33.27	54.17	41.22	41.22	50,453	1,771,306	407
S	38.45	71.83	50.09	50.09	100,810	3,545,420	439
SU20	38.45	71.83	50.09	50.09	100,810	3,545,420	439
SU30	34.66	61.00	44.20	44.20	76,736	2,697,430	436
SU40	37.96	67.50	48.59	48.59	64,488	2,264,004	407
Rao-bGate	34.45	60.17	43.81	43.81	50,105	1,733,448	462
Rao-bS40	35.90	63.00	45.74	45.74	65,510	2,282,729	466
Rao-20	37.79	71.67	49.48	49.48	120,666	4,197,340	467
Rao-40	35.96	67.00	46.80	46.80	106,051	3,674,796	470
Rao-60	33.92	60.67	43.51	43.51	156,591	5,466,736	468
Var-bGate	31.34	53.33	39.48	39.48	52,795	1,845,736	450
Var-bS40	34.71	60.17	44.02	44.02	63,391	2,206,528	454
Var-20	37.75	71.67	49.45	49.45	148,887	5,198,848	462
Var-40	35.76	66.33	46.47	46.47	119,476	4,154,304	468
Var-60	34.21	60.67	43.75	43.75	108,040	3,747,812	459